

More Complicated Logarithmic Equations (Part III)

Solve.

$$\textcircled{1} 3\log_x 2 + \log_x 6 = 1 \quad \boxed{x=48}$$

$$\textcircled{2} 2 = \log_y 8 - 2\log_y 2 \quad \boxed{y=\sqrt{2}}$$

$$\textcircled{3} 4\log_b 2 + \log_b 4 = 3 \quad \boxed{b=4}$$

$$\textcircled{4} 0 = \log_3(4x+1) - \log_3(x+2) \quad \boxed{x=\frac{1}{3}}$$

$$\textcircled{5} \log_5(2-x) - \log_5(x+3) = 1 \quad \boxed{x=-\frac{13}{6}}$$

$$\textcircled{6} 2 = \log_4 Y - \log_4(Y-5) \quad \boxed{y=\frac{16}{3}}$$

★ Hint:
You must
use E VIII
and EX.

$$\textcircled{7} \frac{\log_4 X + 1}{2 + \log_4 X} = 3 \quad \boxed{x=\frac{1}{32}}$$

$$\textcircled{8} 1 = \frac{2 - \log_3 b}{4 - 5\log_3 b} \quad \boxed{b=\sqrt{3}}$$

$$\textcircled{9} -2 = \frac{3\log_2 Y - 30}{2\log_2 Y + 1} \quad \boxed{y=16}$$

$$\textcircled{10} 2^{3+\log_2 X} = 7 \quad \boxed{x=\frac{7}{8}}$$

$$\textcircled{11} 5^{\log_5 Y - 2} = 4 \quad \boxed{y=100}$$

$$\textcircled{12} 6 = 8^{1+\log_8 b} \quad \boxed{b=\frac{3}{4}}$$

$$\textcircled{13} 2 = 12^{\log_{12} X - 1} \quad \boxed{x=24}$$

$$\textcircled{14} \log_2(4x) = \log_2(x+2) \quad \boxed{x=\frac{2}{3}}$$

$$\textcircled{15} 3\log_5 Y = \log_5 2 \quad \boxed{y=\sqrt[3]{2}}$$

★ Do not
forget to
check
for
extraneous
solutions.

$$\textcircled{16} \log_{16}(2b-1) = \log_{16}(3b+7) \quad \boxed{\text{No Solution}}$$

$$\textcircled{17} \log_7 X = 4\log_7 3 \quad \boxed{x=81}$$

$$\textcircled{18} \log_9 2 + \log_9 5 = 2\log_9(3x) \quad \boxed{x=\frac{\sqrt{10}}{3}}$$

$$\textcircled{19} 2\log_5(2x+3) + \log_5 4 = \log_5 16 \quad \boxed{x=-\frac{1}{2}}$$

$$\textcircled{20} 4\log_2(2b) + \log_2 5 = \log_2 240 \quad \boxed{b=\sqrt[4]{3}}$$

$$\textcircled{21} 2 = \log_3[6 + \log_2 X] \quad \boxed{x=8}$$

Warning: The
"2" is not
in the
logarithm.

$$(22) \log_5 [\log_4 Y - 2] = 0$$

$$Y = 64$$

$$(23) 1 = \log_6 [3 + \log_3 (2x - 5)]$$

$$X = 16$$